

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed **end-to-end platform architectures** for robotics systems, integrating sensing, compute, connectivity, and motion components to create reusable building blocks.
- Designed and executed **characterization and test automation** for robotics subsystems, generating performance and robustness data to inform architecture choices.
- Built and iterated on **rapid prototypes** encompassing electrical, mechanical, firmware, and software elements to evaluate new robotics concepts and system feasibility.
- Produced **design documentation and integration guides**, detailing system tradeoffs, pitfalls, and recommended patterns to enable product teams.
- Scripted **Python automation** for data analysis and reporting tasks, supporting the characterization and benchmarking of robotics hardware and software components.
- Managed **incident response and SLA adherence** for enterprise Java/.NET applications, resolving P1/P2 issues within strict production environments.

SKILLS

Robotics & Systems Engineering – Robotic Systems Solutions, Platform Architecture, Rapid Prototyping, System Integration, Embedded Firmware, Real-Time Development, Control Loops, Hardware Bring-up, Lab Measurement, Test Automation, Benchmarking, Design Documentation, Enablement

Software Development – Python, Java, Spring Boot, REST APIs, Scripting, Software Architecture, Debugging, Linux

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD, Jenkins, Git

Data Analysis & BI – SQL, PostgreSQL, MySQL, Oracle, Pandas, NumPy, Data Validation, Data Warehousing

Tools & Methodologies – Jira, Confluence, Agile, Scrum, Change Management

PROJECTS

Robotics Platform Concept Validation (HCL Technologies)

- Developed a proof-of-concept **robotic arm control system**, integrating sensor feedback with custom motion planning algorithms to achieve repeatable positioning within 2mm tolerance.
- Implemented **Python scripting** to automate the collection and initial analysis of sensor data from prototype subsystems, reducing manual data collation time by ~40%.
- Authored a **technical design document** detailing the system architecture, component tradeoffs, and proposed integration strategy for a next-generation autonomous platform.

Embedded Systems Test Automation Framework (HCL Technologies)

- Engineered a **CI/CD pipeline using Jenkins and Docker** to automate the build, test, and deployment of firmware for embedded control modules.
- Developed **Python-based test scripts** that executed functional and performance tests on the firmware, reporting results and identifying regressions across 150+ test cases.
- Integrated **Splunk for logging and monitoring** of test execution, enabling faster root cause analysis for failures in the automated test environment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)